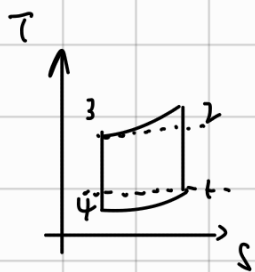


7-2

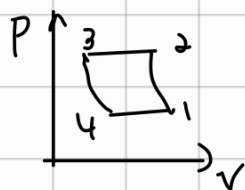


$$\varepsilon = \frac{q_2}{W} = \frac{C_p(T_1 - T_4)}{C_p(T_2 - T_3) - C_p(T_1 - T_4)} = \frac{T_1 - T_4}{T_2 - T_3 - T_1 + T_4} = \frac{1}{\frac{T_2 - T_3}{T_1 - T_4} - 1}$$

$$\text{空气} \Rightarrow K = 1.4$$

$$R_g = \frac{R}{28.97} = 0.287 \text{ kJ/kg} \cdot \text{K}$$

$$C_p = \frac{K R_g}{K - 1} = 1.0045 \text{ kJ/kg} \cdot \text{K}$$



等熵过程: ① - ② 和 ③ - ④

$$\frac{T_2}{T_1} \left(\frac{P_2}{P_1} \right)^{\frac{K-1}{K}} = \left(\frac{P_3}{P_4} \right)^{\frac{K-1}{K}} = \frac{T_3}{T_4}$$

$$\Rightarrow \frac{T_4}{T_1} = \frac{T_3}{T_2}$$

$$= \frac{1}{\frac{T_2(1 - \frac{T_3}{T_2})}{T_1(1 - \frac{T_4}{T_1})} - 1} = \frac{1}{\frac{T_2}{T_1} - 1} = \frac{1}{\left(\frac{P_2}{P_1} \right)^{\frac{K-1}{K}} - 1} = \frac{1}{\left(\frac{0.5}{0.1} \right)^{\frac{1.4-1}{1.4}} - 1}$$

$$= 1.7129$$

制冷量 $Q_2 = 1 \cdot q_2 = C_p(T_1 - T_4)$

$$\frac{T_4}{T_3} = \left(\frac{P_4}{P_3} \right)^{\frac{K-1}{K}} \quad T_4 = T_3 \left(\frac{P_4}{P_3} \right)^{\frac{K-1}{K}} = \left(\frac{P_1}{P_2} \right)^{\frac{K-1}{K}} T_3 = \left(\frac{0.1}{0.5} \right)^{\frac{1.4-1}{1.4}} (288) = 181.8389 \text{ K}$$

$$T_1 = -10 + 273 = 263 \text{ K}$$

$$Q_2 = 1.0045(263 - 181.8389) = 81.5263 \text{ kJ}$$

$$W = \frac{Q_2}{\varepsilon} = \frac{81.5263 \times 10^3}{1.7129} = 47.5955 \text{ kJ}$$

7-3

$$\varepsilon = \frac{1}{\frac{T_2}{T_1} - 1} = 4$$

$$\frac{T_3}{T_1} = \frac{5}{4}$$

$$\varepsilon = \frac{Q}{W}$$

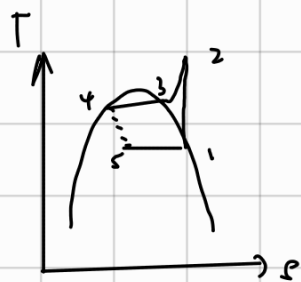
$$Q = W \cdot \varepsilon = 6 \times 10^3 \times 4 = 24 \text{ kW}$$

$$\frac{4T_3}{T_1} - 4 = 1$$

$$\varepsilon' = 1 + \varepsilon = 5$$

$$Q_1 = W \cdot \varepsilon' = 30 \text{ kW}$$

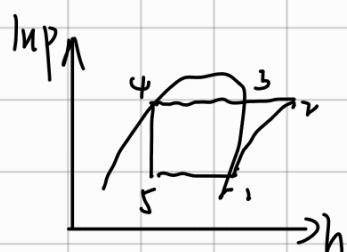
7-7



$$\varepsilon = \frac{q_2}{w} = \frac{h_1 - h_4}{h_2 - h_1 + h_4} = \frac{h_1 - h_4}{h_2 - h_1}$$

$$T_5 = -20^\circ\text{C} = T_1 \quad T_3 = 30^\circ\text{C}$$

$$\varepsilon = \frac{q}{w}$$



$$h_1(-20^\circ\text{C}) = 385.89 \text{ kJ/kg}$$

$$h_4(30^\circ\text{C}) = 241.80 \text{ kJ/kg}$$

$$p_2 = p_3 = 770.21 \text{ kPa}$$

$$s_2 = s_1 = 1.7387 \text{ kJ/kg}\cdot\text{K}$$

$$p = 0.7 \text{ MPa} \quad s = 1.727 \quad h = 416.37 \text{ kJ/kg}$$

$$p = 0.7 \text{ MPa} \quad s = 1.7593 \quad h = 426.32 \text{ kJ/kg}$$

$$p = 0.7 \text{ MPa} \quad s = 1.7387 \quad h = 416.37 + \frac{1.7387 - 1.727}{1.7593 - 1.727} (9.95) = 419.9742 \text{ kJ/kg}$$

$$p = 0.8 \text{ MPa}, \quad s = 1.7435, \quad h = 424.31 \text{ kJ/kg}$$

$$p = 0.8 \text{ MPa} \quad s = 1.7753 \quad h = 424.41 \text{ kJ/kg}$$

$$p = 0.8 \text{ MPa} \quad s = 1.7387 \quad h = 424.31 - \frac{1.7435 - 1.7387}{1.7753 - 1.7435} (10.1) = 422.684 \text{ kJ/kg}$$

$$p = 0.7702 \text{ MPa} \quad s = 1.7387 \text{ kJ/kg}\cdot\text{K} \quad h_2 = 419.9742 + \frac{0.7702 - 0.7}{0.8 - 0.7} (2.7198)$$

$$h_2 = 421.8835 \text{ kJ/kg}$$

$$\varepsilon = \frac{h_1 - h_4}{h_2 - h_1} = 4.0032 \approx 4$$

$$1 \text{ kW} = m q_2 = m (h_1 - h_4) \quad \dot{m} = \frac{1}{h_1 - h_4} = 6.9401 \times 10^{-3} \text{ kg/s}$$

$$\dot{w} = \dot{m} (h_2 - h_1) = 0.2498 \text{ kW}$$